

Milestone 5. Deployment

Activity 5.1. Preparing the Application for Local Deployment

Activity 5.2. Testing and Verifying Local Deployment

Milestone 5. Deployment

Now, the focus is on deploying the FitBuddy application on a local system using FastAPI and Uvicorn. The app uses HTML templates rendered with Jinja2, alongside AI services powered by Gemini. This milestone ensures all environment dependencies are correctly installed, and the app runs smoothly on a local server, preparing it for future deployment on cloud platforms.

Activity 5.1: Preparing the Application for Local Deployment

Set up a virtual environment

Create a virtual environment to manage dependencies and isolate the FitBuddy project from other environments.

Activate the environment and install all necessary dependencies listed in the requirements.txt file, including FastAPI, Uvicorn, SQLAlchemy, and Google Generative AI SDK.

```
Python -m venv venv
```

```
Source venv/bin/activate # On Windows: venv\Scripts\activate
```

```
Pip install -r requirements.txt
```

Configure environment variables

Use environment variables to store sensitive data such as the Gemini API Key.

These variables can be set directly in the terminal or added to a .env file which is loaded into the application using python-dotenv.

```
GOOGLE_API_KEY=your_gemini_api_key_here
```

Frontend-Backend Rendering via Jinja2

All user interactions are facilitated via Jinja2-rendered templates:

Index.html: Collects user inputs

Result.html: Displays the generated AI plan

All_users.html: Lists all user records for admin view

Activity 5.2: Testing and Verifying Local Deployment

Access the Application Locally

The FastAPI development server must be started using the command (uvicorn main:app — reload). Once the server is running, the application can be accessed through any web browser by navigating to <http://127.0.0.1:8000>. This URL loads the main user interface where individuals can input their details and generate personalized fitness plans.

Additionally, the interactive API documentation provided by FastAPI can be viewed at <http://127.0.0.1:8000/docs>, allowing developers to explore and test available endpoints.

Exploring Website's web pages:

Home page:

FitBuddy – AI Workout Generator

Name:

User ID:

Age:

Weight (kg):

Fitness Goal:

e.g. weight loss, flexibility

Workout Intensity:

Low

Description: The homepage of FitBuddy – AI Workout Generator features a clean and engaging user interface designed to provide a seamless experience for users seeking personalized fitness plans. The form prominently displayed in the center allows users to enter essential information including:

Name

User ID

Age

Weight (kg)

Fitness Goal (e.g., weight loss, flexibility)

Workout Intensity (Low, Medium, High)

Once users submit this data, they can click the “Generate Plan” button, which triggers the AI-powered backend to generate a customized workout and nutrition plan tailored to their goals and fitness profile. The visual appeal of the background image featuring an active individual reinforces the fitness-centric theme of the application, motivating users to take charge of their health journey.

This intuitive design ensures users of all experience levels can quickly get started with minimal effort.

Personalized workout generation page:

User Information:

Generate Plan



Your Personalized Workout Plan

User Information

Name: xyz

User ID: 10

Age: 20

Weight: 70.0 kg

Goal: i want to lose belly fat and gain muscles

Intensity: High

Description: Once the user submits their details, the application navigates to a personalized workout plan page. This section displays a summary of the user's input including their Name, User ID, Age, Weight, Fitness Goal, and Selected Intensity. This personalized display ensures users feel acknowledged and guides the AI engine to generate relevant workout recommendations tailored specifically to their fitness journey.

Workout Plan:

Workout Plan

7-Day High-Intensity Workout Plan for Fat Loss & Muscle Gain

This plan focuses on compound exercises to maximize calorie burn and muscle engagement. Remember to adjust the intensity based on your fitness level and consult a doctor before starting any new workout routine. Proper nutrition is essential for achieving your goals, so ensure you're supporting your training with a healthy diet.

****Day 1: Upper Body Strength****

****Warm-up (5 mins):**** Jumping jacks (60 seconds), high knees (30 seconds), arm circles (forward and backward, 30 seconds each), dynamic stretches like arm swings and torso twists (1 min).

****Main Workout:****

- ****Barbell Bench Press:**** 3 sets of 8-12 reps
- ****Pull-ups (or Lat Pulldowns):**** 3 sets of 8-12 reps
- ****Overhead Press:**** 3 sets of 8-12 reps
- ****Barbell Row:**** 3 sets of 8-12 reps
- ****Dumbbell Bicep Curls:**** 3 sets of 10-15 reps
- ****Dumbbell Triceps Extensions:**** 3 sets of 10-15 reps

****Cooldown:**** 5 min static stretches holding each for 20 seconds (chest, back, biceps, triceps, shoulders).

****Day 2: Lower Body & Core****

****Warm-up (5 mins):**** Bodyweight squats (15 reps), lunges (10 reps per leg), glute bridges (15 reps), plank (30 seconds).

****Main Workout:****

- ****Barbell Squats:**** 3 sets of 8-12 reps
- ****Romanian Deadlifts:**** 3 sets of 8-12 reps
- ****Walking Lunges:**** 3 sets of 12-15 reps per leg
- ****Glute Bridges:**** 3 sets of 15-20 reps
- ****Hanging Leg Raises:**** 3 sets to failure
- ****Russian Twists:**** 3 sets of 15-20 reps per side

****Cooldown:**** 5 min cool down, focusing on quads, hamstrings, and glutes. Static stretches for hip flexors, hamstrings, and glutes (30 seconds each).

****Day 3: HIIT Cardio & Core****

****Warm-up (5 mins):**** Light cardio, like jogging or jumping jacks, followed by dynamic stretches.

****Main Workout:****

- ****Burpees:**** 4 sets of 10-15 reps
- ****Mountain Climbers:**** 4 sets of 30-60 seconds
- ****Jump Squats:**** 4 sets of 10-15 reps
- ****Kettlebell Swings:**** 4 sets of 15-20 reps
- ****Plank variations (high plank, forearm plank, side plank):**** 30-60 seconds each, repeat 2-3 times

****Cooldown:**** 5 min light cool down, static stretches for core and legs.

****Day 4: Rest or Active Recovery****

****Active Recovery:**** Light activity like walking, swimming, yoga, or foam rolling. Focus on mobility and flexibility. This helps promote blood flow and reduces muscle soreness.

****Day 5: Upper Body Strength (focus on different exercises)****

****Warm-up (5 mins):**** Similar to Day 1.

****Main Workout:****

- ****Incline Dumbbell Press:**** 4 sets of 8-12 reps
- ****Chin-ups (or Close Grip Lat Pulldown):**** 4 sets of 8-12 reps
- ****Arnold Press:**** 3 sets of 8-12 reps
- ****T-Bar Row:**** 3 sets of 8-12 reps
- ****Hammer Curls:**** 3 sets of 10-15 reps
- ****Overhead Triceps Extensions:**** 3 sets of 10-15 reps

****Cooldown:**** Similar to Day 1.

****Day 6: Lower Body & Core (Focus on different exercises)****

****Warm-up (5 mins):**** Similar to Day 2.

****Main Workout:****

- ****Front Squat:**** 3 sets of 8-12 reps
- ****Good Mornings:**** 3 sets of 10-12 reps
- ****Bulgarian Split Squat:**** 3 sets of 10-12 reps per leg
- ****Step-ups:**** 3 sets of 12-15 reps
- ****Cable Crunches:**** 3 sets of 15-20 reps
- ****Wood Chops (cable rotation):**** 3 sets of 12-20 reps per side

****Cooldown:**** Similar to Day 2.

****Day 7: Rest or Active Recovery****

****Active Recovery:**** Similar to Day 4. Prioritize getting enough sleep this day to prepare for the next week of training.

****Important Notes:****

- ****Progressive Overload:**** Gradually increase the weight, reps, or sets each week to challenge your muscles and promote continued growth.
- ****Proper Form:**** Focus on maintaining correct form throughout each exercise to prevent injury and maximize results. Watch videos and, if possible, consult with a trainer to ensure proper technique.
- ****Listen to your Body:**** Rest when needed and don't push through pain. Adjust the plan as needed based on your recovery and progress.
- ****Nutrition:**** Fuel your body with a balanced diet rich in protein, complex carbohydrates, and healthy fats to support muscle growth and recovery.
- ****Hydration:**** Drink plenty of water throughout the day, especially before, during, and after workouts.

This plan is a starting point. You can adjust it based on your progress and preferences. Remember consistency and proper execution are key to achieving your fitness goals. Good luck!

Description: FitBuddy generates a personalized, day-wise workout plan tailored to the user's fitness goal, experience level, and preferred workout schedule. The workout section on the result page outlines each day with a clear focus, such as full body, upper or lower body, cardio, core strength, or flexibility. For every workout day, a curated list of exercises is provided, detailing the name of the exercise, the number of sets and repetitions (or duration), and suggested rest intervals. The plan incorporates variety and structured progression to keep the workouts engaging and effective. Additionally, basic warm-up and cool-down guidance may be included to support safe and well-rounded training. Overall, this section aims to deliver a goal-oriented, easy-to-follow workout schedule that promotes consistency and progress.

Nutrition Tip:

Nutrition Tip

Prioritize protein! Aim for a good source of protein (like chicken, fish, beans, or Greek yogurt) with every meal. Protein helps build muscle, keeps you feeling full, and supports your metabolism, all crucial for losing belly fat and gaining muscle.

Description: The Nutrition Tip section in FitBuddy provides users with concise, practical dietary advice to complement their workout plans. It emphasizes the importance of prioritizing protein intake in every meal, suggesting sources like chicken, fish, beans, or Greek yogurt. The tip explains how protein contributes to muscle building, satiety, and metabolic support—key factors for both fat loss and muscle gain. This section is designed to reinforce healthy eating habits that align with users' fitness goals.

Feedback Page:

Share Your Feedback

Your Unique User ID:

Enter the same User ID used to generate your plan

Your Feedback:

Let us know how we can improve your plan...

Submit Feedback

Description: The Feedback Page in FitBuddy allows users to personalize and refine their fitness journey by submitting feedback on their current workout plans. Users are prompted to enter their unique User ID and provide specific feedback regarding the effectiveness, difficulty, or preferences related to their plan. Once submitted, this feedback is processed

by the system to intelligently update and adjust the workout plan to better align with the user's goals, comfort, and progress. This dynamic feedback loop ensures that each fitness plan remains adaptive and tailored to the individual's evolving needs.

View all users page:

FitBuddy – All Users & Workout Plans

User ID: 1

Name: Shreya

Age: 22

Weight: 55.0 kg

Goal: Muscle gain

Intensity: High

Original Plan: Day 1: Upper Body Push

Warm-up

Arm circles (forward and backward, 2 x each)

Shoulder push-ups (1 min)

Push-up variations (as many reps as possible in 1 min)

Light bench press (empty bar, 1 set of 10–12 reps)

Main workout:

Incline Press

Shoulder Press

Dumbbell Shoulder Press

Overhead Press

Triceps Pushdown

Lateral Raise

Day 2: Lower Body (Legs & Core)

Squats

Romanian deadlift

Lunges

Leg extensions

Plank

Updated Plan: Day 1: Upper Body Push

Warm-up

Arm circles (forward and backward, 2 x each)

Shoulder push-ups (1 min)

Push-up variations

Light bench press

Main workout:

Incline Press

Shoulder Press

Dumbbell Shoulder Press

Overhead Press

Triceps Pushdown

Lateral Raise

Description: The View All Users page in FitBuddy provides an overview of the entire user database, displaying detailed information for each registered user. This includes their personal details such as name, age, and fitness goals, along with their original workout plan generated at registration and the updated workout plan if any changes were made based on their submitted feedback. This feature enables a transparent view of how each user's plan has evolved over time, showcasing the adaptability of the FitBuddy system and helping administrators or developers monitor user progress and engagement effectively.

Conclusion:

The FitBuddy – AI Fitness Plan Generator is a comprehensive, user-centric web application that showcases the powerful synergy between artificial intelligence and health-tech solutions. Designed to assist individuals in achieving their fitness goals, FitBuddy uses Google's Gemini 1.5 Pro and Gemini Flash models to generate personalized 7-day workout plans and daily nutrition or recovery tips based on user inputs such as age, weight, fitness goal, and preferred workout intensity.

The application is built using the FastAPI framework for backend logic and API handling, SQLAlchemy with SQLite for persistent data storage, and Jinja2 templating for a clean and responsive frontend. Upon submitting their details on the homepage, users receive a tailored workout plan that includes warm-ups, main workouts, cooldowns, and goal-aligned tips. What sets FitBuddy apart is its dynamic feedback feature—users can provide input on the effectiveness or difficulty of their current plan, and the system responds by intelligently updating the plan using generative AI, ensuring ongoing customization and engagement.

Additionally, the application includes an administrative view where all registered users and their respective plans (original and updated) are displayed in a structured format. This functionality supports transparency, monitoring, and further optimization, making FitBuddy suitable for both individual users and coaches or institutions managing multiple fitness profiles.

Overall, FitBuddy stands as a scalable and adaptable AI-powered solution that not only personalizes the fitness journey but also evolves with the user's progress and preferences. It demonstrates how cutting-edge technology can be integrated into everyday wellness routines to promote healthy, consistent, and goal-oriented lifestyles.